Task 1

Cut out the statements and put them in the correct place in the table to show how different scientists contributed to the development of the DNA model.

Francis Crick, James Watson and Maurice Wilkins received the Nobel Prize for Medicine for their work on the structure of DNA.

Rosalind Franklin (born in 1920) attended one of the few secondary schools in London where girls were taught science.

X-ray crystallography was used to find the position of atoms in simple molecules, the first structure to be determined was that of table salt.

Rosalind Franklin had become an expert in the technique of x-ray crystallography. The famous image ‘Photo 51’ was taken.

Wilkins completed seven years of work in order to check and verify Crick and Watson’s model of the DNA molecule.

Rosalind Franklin died of cancer at the age of 37.

Crick and Watson made numerous models of the DNA molecule; first out of cardboard and then of metal plates and brass rods.

Wilkins shows ‘Photo 51’ to Crick and Watson without Rosalind Franklin’s permission.

Linus Pauling and Maurice Wilkins turned away from studying nuclear science and began to study biological molecules.

Modern NMR spectrometers can show biological proteins interacting with other molecules.

Crick and Watson revealed their double helix model of DNA.

Atomic bombs were dropped on Hiroshima and Nagasaki in Japan with devastating effects.

Oswald Avery proved that DNA, and not the protein the part of chromosomes, was responsible for inheritance.

1914	
1934	
1940s	Erwin Chargaff found that the number of A and T bases were the same, as were the number of C and G bases.
1942	
1945	
1946	
1950	Wilkins took the first ever images of the DNA molecule.
1951	Linus Pauling discovered that the molecules of some proteins have a helical shape.
1952	
1952-3	
1953	
1953	
1958	
1960	
1962	
1990s	

Task 2

Use the completed table to answer the following questions.

1. Crick and Watson revealed their double helix model of DNA in 1953. Why was the Nobel Prize for Medicine not awarded to them until 1962?

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2. 'Photo 51' provided Crick and Watson with crucial information about the structure of DNA. Why do you think Rosalind Franklin was not awarded a Nobel Prize, along with Crick, Watson and Wilkins in 1962?

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3. Pauling and Wilkins began their careers researching the structure of atoms. What do you think made them change to studying biological molecules instead?

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Task 3

- Cut out the picture pairs.
- Use the table to find the names of the seven different scientists.
- Match the picture pairs to the seven scientists involved in the development of the DNA model. Write the names of the scientists onto the correct pictures.

Double helix snakes and ladders board

Finish: 1990s	1962	1960 Wilkins work checking Crick and Watson's model of DNA was completed	1958
1951 Linus Pauling discovered that some protein molecules have a helical shape	1952	1952-3	1953
1951	1950 Wilkins took the first ever images of the DNA molecule	1946	1945
Start: 1914	1934	1940s Erwin Chargaff found that bases A = T and C = G	1942

Double helix snakes and ladders

Instructions

- Cut up the statements and put them into the correct place on the snakes and ladders board.
- Add you own ladders of base pairs and double helical snakes.
- Add instructions to three squares such as ‘move forward 3 spaces’ or ‘go back to the start’.
- Each choose one of the scientists and make yourself a counter with their name.
- Play the game.

X-ray crystallography used to find the position of atoms in a crystal of table salt.	Rosalind Franklin (born in 1920) attended a school where girls were taught science.	Oswald Avery proved that DNA, is responsible for inheritance.
Atomic bombs destroyed Hiroshima and Nagasaki, Japan.	Linus Pauling and Maurice Wilkins began to study biological molecules	Crick and Watson revealed their double helix model of DNA.
The famous image ‘Photo 51’ was produced.	Wilkins showed ‘Photo 51’ to Crick and Watson.	Crick and Watson made numerous models of the DNA molecule.
Rosalind Franklin died of cancer at the age of 37.	Modern spectrometers showed biological proteins interacting together.	Crick, Watson and Wilkins received the Nobel Prize for Medicine.



Teaching notes

This resource has 16 ‘milestones’ in the study of the DNA molecule. Included is the work of Crick, Franklin, Watson and Wilkins. Also included are important contributions by other scientists.

The information can be used in several other ways.

- Enlarge the statements and give one to each pair of students. Ask the students to add them to a ‘washing line’ timeline. Other major events of the century could also be added.
- Ask the students to find the names of the seven scientists in the statements. They could then group the statements into piles according to the scientists. Ask students to write a summary of each scientist’s contribution to the development of the DNA model.
- Ask students to draw a concept map to show the links between the scientists, their work and the development of the DNA model.
- Working in pairs, students write a short script in which the scientists explain their contribution towards the development of the DNA model. They could make paper finger (or pencil) puppets of each of the scientists using the photos (they could try to find a photo of Erwin Chargaff) and perform their script. They could video their presentations to show to the class later.

Snakes and ladders teaching notes

To give students a better understanding of the timescale of this work give students A3 pieces of paper and ask them to draw their own game board, 10 x 10 squares, starting around 1900 and ending around the year 2000 (they will need to allow three squares for 1951).

Answers

Task 1

1914 X-ray crystallography was used to find the position of atoms in simple molecules, the first structure to be determined was that of table salt.
1934 Rosalind Franklin attended one of the few schools in London where girls were taught science.
1940s Erwin Chargaff found that the number of A and T bases were the same, as were the number of C and G bases.
1942 Oswald Avery proved that DNA, and not the protein part of chromosomes, was responsible for inheritance.
1945 Atomic bombs are dropped on Hiroshima and Nagasaki in Japan with devastating effects.
1946 Linus Pauling and Maurice Wilkins turned away from studying nuclear science and began to study biological molecules.
1950 Wilkins took the first ever images of the DNA molecule.
1951 Linus Pauling discovered that the molecules of some proteins have a helical shape.
1952 Rosalind Franklin had become an expert in the technique of x-ray crystallography. The famous image ‘Photo 51’ was taken *(by Gosling a research scientist, supervised by Franklin).*

1953 Wilkins showed ‘Photo 51’ to Crick and Watson without Rosalind Franklin’s permission.
1952-3 Crick and Watson made numerous models of the DNA molecule; first out of cardboard and then of metal plates and brass rods.
1953 Crick and Watson revealed their double helix model of DNA.
1958 Rosalind Franklin died of cancer at the age of 37.
1960 Wilkins completed seven years of work in order to check and verify Crick and Watson’s model of the DNA molecule.
1962 Francis Crick, James Watson and Maurice Wilkins received the Nobel Prize for Medicine for their work on the structure of DNA.
1990s Modern NMR spectrometers can show biological proteins interacting with other molecules.

Task 2

1. Crick and Watson revealed their double helix model of DNA in 1953. Why was the Nobel Prize for Medicine not awarded to them until 1962?

   The Nobel Prize was not awarded until 1962 because it took other scientists, most notably, Maurice Wilkins seven years to check and verify the work of Crick and Watson.

2. ‘Photo 51’ provided Crick and Watson with crucial information about the structure of DNA. Why do you think Rosalind Franklin was not awarded a Nobel Prize, along with Crick, Watson and Wilkins in 1962?

   She was not awarded the Nobel Prize as it is not awarded posthumously and Rosalind Franklin died at the age of 37 in 1958.

3. Pauling and Wilkins began their careers researching the structure of atoms. What do you think made them change to studying biological molecules instead?

   They moved away from studying atoms after atomic bombs were dropped on Hiroshima and Nagasaki. They were concerned about the further development of nuclear weapons, developments which involved research by many scientists. Indeed, Linus Pauling became a prominent peace activist and was awarded the Nobel Peace Prize in 1962.

Task 3

1. Rosalind Franklin and ‘Photo 51’
2. Watson and Crick (Watson on left) and DNA molecule.
3. Linus Pauling and a helical protein (collagen).
4. ‘Erwin Chargaff’ and paired bases.
5. Oswald Avery and cell, nucleus, chromosome, DNA to represent his discovery that DNA is responsible for inheritance.
6. Maurice Wilkins and an x-ray diffraction camera from the 1950s.